

ICP Final Project Report

Project Title: Supermarket Inventory Management System

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Submission Date: August 11, 2026

1. Abstract

The Supermarket Inventory Management System is a basic C programming project designed to store and manage product information in a supermarket. The system uses arrays to store product IDs, prices, and quantities. It allows the user to add products, display all stored products, search for a product using its ID, and sort products according to price. The project demonstrates fundamental programming concepts such as arrays, loops, conditional statements, linear searching, and sorting. The system is menu-driven so that the user can repeatedly choose an operation until selecting the exit option.

2. Introduction

Managing products manually can become difficult when the number of items increases. An inventory system provides a simple way to keep track of products, quantities, and prices. The main objective of this project is to create a beginner-friendly supermarket inventory program in C using basic programming concepts. The final simplified version does not use structures or user-defined functions.

The program stores up to 50 products. Three parallel arrays are used: one for product IDs, one for prices, and one for quantities. The same array index represents the same product across all three arrays.

3. Methodology / Implementation

The program begins by declaring `id[50]`, `price[50]`, and `quantity[50]`. A variable `n` keeps track of the number of products currently stored. A do-while loop repeatedly displays the menu and processes the user's choice.

When a product is added, its ID, price, and quantity are stored at position `n`, and `n` is increased by one. To display products, a for loop travels through the arrays and prints each item. Searching uses linear search: each product ID is checked one by one until the required ID is found or the list ends.

Sorting compares the prices of two products. If the earlier price is greater than the later price, the ID, price, and quantity are swapped together. Repeating these comparisons arranges the products from the lowest price to the highest price.

Sample Code Visual

Code Visual - Add and Search

```
/* Add Product */
printf("Enter Product ID: ");
scanf("%d", &id[n]);

printf("Enter Price: ");
scanf("%f", &price[n]);

printf("Enter Quantity: ");
scanf("%d", &quantity[n]);

n++;

/* Search */
for(i = 0; i < n; i++)
{
    if(id[i] == searchID)
    {
        printf("Product Found!");
        break;
    }
}
```

Figure 1. Example code used for adding and searching products.

4. Results / Output

The completed program provides a simple menu through which the user can perform inventory operations. In a sample run, three products are entered, displayed, searched by ID, and then sorted by price.

```
Sample Input / Output
===== SUPERMARKET INVENTORY =====
1. Add Product
2. Display Products
3. Search Product
4. Sort by Price
5. Exit

ID   Price   Quantity
1    80.00   10
2    30.00   5
3    50.00   8

Search ID: 2
Product Found!
ID: 2
Price: 30.00
Quantity: 5

After sorting by price:
2    30.00   5
3    50.00   8
1    80.00   10
```

Figure 2. Sample input and output of the inventory system.

5. Challenges and Limitations

One challenge was keeping the product information synchronized while sorting. Because the project uses separate arrays, the ID, price, and quantity must all be swapped together. Another limitation is that the program stores only 50 products and does not save information after the program closes. The simplified version also does not include product names, updates, or deletion.

6. Conclusion and Discussion

The Supermarket Inventory Management System successfully demonstrates how basic C programming concepts can be applied to a real-world problem. The program can add, display, search, and sort inventory items. Arrays and loops provide a simple method of organizing product information, while searching and sorting make the stored data easier to use.

7. Future Work

The system can be improved by adding product names, updating and deleting products, checking for duplicate IDs, validating prices and quantities, and saving inventory data to a file. A future version could also use structures and separate functions to make the program more organized.

8. References

1. Kernighan, B. W., & Ritchie, D. M. (1988). The C Programming Language (2nd ed.). Prentice Hall.
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9. Individual Contributions

Marufa:Contributed to the planning and design of the supermarket inventory system and the organization of the project report.

Junaira : Contributed to the C program implementation, including array-based data storage, the menu system, product searching, price sorting, testing, and preparation of the project report.

Both members participated in reviewing the final project, checking the program flow, and discussing possible improvements.